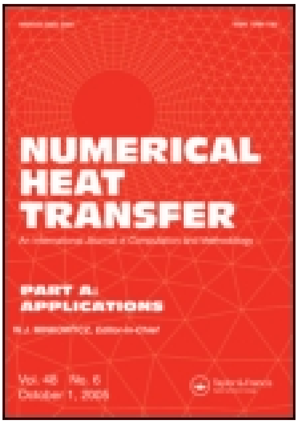

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Heat Transfer Analysis of Functionally Graded Hollow Cylinders Subjected to an Axisymmetric Moving Boundary Heat Flux

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HEAT TRANSFER ANALYSIS OF FUNCTIONALLY GRADED HOLLOW CYLINDERS SUBJECTED TO AN AXISYMMETRIC MOVING BOUNDARY HEAT FLUX

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The transient heat transfer analysis of functionally graded (FG) hollow cylinders subjected to a distributed heat flux with a moving front boundary on its inner surface is presented. The heat flux is assumed to be axisymmetric, and its front boundary moves along the axis of the cylinder. A method composed of the finite element and differential quadrature methods is employed to discretize the governing equations in the spatial domain. After demonstrating the convergence and accuracy of the method, the effects of different parameters on the temperature distribution and time history of the temperature at different points of FG cylinder are investigated.

1. INTRODUCTION

Functionally graded materials (FGMs) are nonhomogeneous materials within which their thermomechanical properties have smooth and continuous spatial variation due to a continuous change in composition, morphology, microstructure, or in crystal structure. FGMs can take advantage of desirable properties such as heat and corrosion resistance of ceramics and high tensile strength, toughness, and bonding capability of metals. These materials have been developed for use in high temperature environments such as aerospace structures, reactors, and other engineering fields. Usually, the structural elements made of FGMs such as beams, plates, and shells are subjected to a high temperature environment which induces thermal stresses within them. Hence, a precise evaluation of their thermal characteristic is of great interest for engineering design and manufacturing [1–10].

Most heat conduction problems are described and analyzed using Fourier's and Fick's laws. It is well known that these classical diffusion theories may break down when one is interested in the transient problems in an extremely short period of time, very high flux, or temperatures near absolute zero [11–23]. Then, the non-Fourier heat conduction becomes trustworthy in describing the diffusion process and predicting the temperature distribution. To better explain heat conduction, non-Fourier heat conduction theories have been developed. One of the non-Fourier

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NOMENCLATURE

A_{ij}^{r}	first-order weighting coefficient	t	time
B_{ij}^{r}	second-order weighting coefficient	T	temperature distribution
c	specific heat capacity	T_0	initial temperature
h	thickness of the hollow cylinder	T_∞	ambient temperature
h_c	convective heat transfer coefficient	$\hat{T}$	vector of temperature
K	thermal conductivity	T^*	nondimensional temperature
L	length of the hollow cylinder	$\overline{T}$	transient nondimensional temperature
N_e	number of elements along the z-direction	u	velocity of heat flux
N_r	grid points along the r-direction	V_e	vernotte number
N_z	total number of nodes along the z-direction	$z_0(t)$	location of heat flux front edge
		α_c	thermal diffusivity of the ceramic
p	power law index	δT	variation of temperature
P_m	material property of the metal	$\varphi_i(z)$	global Lagrange interpolation functions
P_c	material property of the ceramic		
$\vec{q}(r,z,t)$	heat flux vector	τ	relaxation time
$\hat{q}_r$, q_0(t)	prescribed heat flux on the boundary and the heat source strength	η	nondimensional coordinate variable along the z-axis
		ρ	density
q_n	heat flux	$\vec{\nabla}$	two-dimensional gradient vector in a cylindrical coordinate system
r, z	cylindrical coordinate variable		
R_i and R_o	inner and outer radi of the hollow cylinder	ξ	nondimensional coordinate variable along the r-axis

theories is hyperbolic heat conduction theory. This theory, separately proposed by Vernotte [18] and Cattaneo [19], accounts for the finite speed of thermal energy propagation by introducing a new material property called thermal relaxation time.

In comparison with the heat transfer analysis of FG cylinders based on Fourier's law (see for example [1–10].), there are few works which include the effect of a finite value of heat wave speed on heat conduction of FG cylinders [20–22]. In this interesting research [20–22], the analytical solutions for the one-dimensional heat transfer of FG cylinders with constant relaxation time subjected to the distributed stationary heat fluxes, prescribed temperature, or convection on the external/internal surfaces of FG cylinders were presented. On the other hand, the temperature field induced by a moving heat source in hollow FG cylinders are of more concern for many engineering applications like aviation, rocketry, solid propellant rocket motors, gun barrels, nuclear reactor elements, engine exhaust liners, missiles, chemical, aerospace and mechanical industries, etc.

The heat transfer analysis of hollow isotropic and composite cylinders subjected to an axially moving heat source were studied by some researchers [23–28]. Chu et al. [23] presented an analytical solution for the temperature and stress distribution in a finite composite hollow cylinder under a moving line source by applying a hyperbolic heat conduction equation. They employed Laplace transform, eigenfunction expansion, and Fourier series methods to solve the problem.

Chen and Chu [24, 25] studied the transient temperature distribution and thermal stress in a two-layered hollow cylinder subjected to a line heat source which moves axially with a constant velocity [24] or harmonically [25]. They used the

Laplace transform together with the eigenfunction expansion method to solve the problem.

Ootao et al. [1] studied the transient temperature and thermal stress distribution in an infinitely long nonhomogeneous hollow cylinder due to a moving heat source in the axial direction from the inner and/or outer surfaces using the layerwise theory in conjunction with the method of Fourier cosine and Laplace transformations.

Hsieh and Leung [26] solved a quasi-steady state phase-change problem for a cylindrical shell, of which the inner or outer surfaces were heated with an axisymmetric ring heater moving at a constant velocity. After transforming the heat transfer equation to a Klein-Gordon equation, a method of undetermined parameters was employed to express the results in Fredholm integral equations. These equations were solved using the least-square iteration method.

Alzaharnah [27] presented a numerical investigation on the temperature gain and thermal stress distribution developed in a titanium cylinder when heated internally by a constant speed moving ring heat source which had a periodic content.

Jabbari et al. [28] developed an analytical solution for the one-dimensional temperature distribution, mechanical, and thermal stresses in an FG hollow cylinder under a moving heat source, which moves across the thickness of the cylinder.

From the above literature review, it is revealed that the temperature distribution of FG hollow cylinders subjected to a moving heat source based on the non-Fourier heat conduction equation has not yet been investigated. This motivated us to study this problem here. The FG cylinder is subjected to an axisymmetric heat flux on its inner surface which its upper edge moves along the axial direction. The governing equation is discretized along the longitudinal direction using the finite element method (FEM). Then, the differential quadrature method (DQM), as an efficient and accurate numerical tool [17, 29–34], is applied to discretize the strong form of the resulting variable coefficients partial differential equations and the related boundary conditions in the thickness direction. The obtained ordinary differential equations are then solved in the temporal domain using the Newmark time integration scheme [35]. The effects of the material properties, the relaxation time, the velocity of the heat flux moving edge and geometrical parameters on the temperature distribution, and also the time history of temperature at different points of FG cylinder are investigated.

2. GOVERNING EQUATIONS

Consider an FG hollow cylinder which is subjected to a distributed heat flux with moving upper boundary on its inner surface, as shown in Figure 1. Its outer surface is cooled by convection heat transfer. Due to axisymmetric geometry, material properties and thermal loading conditions, the heat conduction equations reduce to two-dimensional ones. Consequently, a cylindrical coordinate system (r, z) is sufficient to label the material points of the cylinder.

The material properties are assumed to vary continuously through the thickness direction according to power law distribution in terms of the volume fractions of the constituents. The material composition continuously varies such that the inner surface of the cylinder is ceramic rich, whereas the outer surface of the cylinder is metal rich. Based on the power law distribution, a typical effective material property

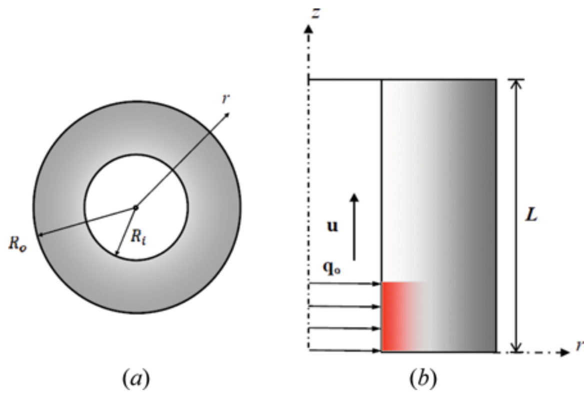

Figure 1. Geometry and thermal loading of an FG hollow cylinder: (a) cross-sectional geometry of the FG cylinder, (b) longitudinal geometry and thermal loading of the FG cylinder (color figure available online).

P of the FG hollow cylinder is obtained as

$$P(r) = P_c + \left(P_m - P_c\right)\left(\frac{r - R_i}{R_o - R_i}\right)^p \tag{1}$$

where the subscripts m and c refer to the metal and ceramic constituents, respectively; p is the power law index (or the material property graded index); and R_i and R_o are the inner and outer radius of the hollow cylinder, respectively. Clearly, when $p = 0$, one obtains $P(r) = P_m$ and when p is very large $P(r)$ inclined to P_c.

The hyperbolic heat transfer constitutive equation and the energy equation without the internal heat generation source can be written as

$$\tau\frac{\partial \vec{q}}{\partial t} + \vec{q} = -K\vec{\nabla}T, \rho c\frac{\partial T}{\partial t} = -\vec{\nabla}.\vec{q} \tag{2a, 2b}$$

where T $[= T(r,\ z,\ t)]$ is the temperature, $K[= K(r)]$ the thermal conductivity, $\rho[= \rho(r)]$ the mass density, $c[= c(r)]$ the specific heat capacity, and $\vec{q}(r, z, t)$ is the heat flux vector at an arbitrary material point of the cylinder, respectively. Also, t is time, τ the relaxation time, and $\vec{\nabla}$ the two-dimensional gradient vector in cylindrical coordinate system.

Eliminating the heat flux vector in Eq. (2), one gets the following.

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\mathrm{K}\frac{\partial T}{\partial r}\right) + \frac{\partial}{\partial z}\left(\mathrm{K}\frac{\partial T}{\partial z}\right) = \rho c\left(\frac{\partial T}{\partial t} + \tau\frac{\partial^2 T}{\partial t^2}\right) \tag{3}$$

Different thermal boundary conditions can be considered on the surfaces of the cylinder. Without loss of generality, the following initial and boundary conditions are assumed in this study.

$$T(r, z, 0) = T_0, \left.\frac{\partial T}{\partial t}\right|_{t=0} = 0 \tag{4a, 4b}$$

$$\text{At } z = 0,\ L : T(r, z, t) = T_0 \tag{5a, 5b}$$

$$\text{At } r = R_i : -K\frac{\partial T}{\partial r} = \hat{q}_r = \begin{cases} q_0(\mathrm{t}) & 0 \leq z \leq z_0(t) \\ 0 & z_0(t) \leq z \leq L \end{cases} \quad \text{at } r = R_o : K\frac{\partial T}{\partial r} = h_c(T_\infty - T) \tag{6a, 6b}$$

where T_0 is the initial temperature of the FG cylinder, which is assumed to be equal to the ambient temperature T_∞; $z_0(t)$ is the location of heat flux front edge; h_c is the convection heat transfer coefficient on the outer surface of the cylinder; and $\hat{q}_r$ and $q_0(t)$ are the prescribed heat flux on the boundary and the heat source strength, respectively. It is assumed that the ends of the FG cylinder are held in contact with large heat capacity environments to keep the temperature at its ends constant.

3. SOLUTION PROCEDURE

Due to complex thermal boundary condition on the inner surface of the cylinder, the finite element method is employed to discretize the governing equations along the axial direction (z-axis).

For this purpose, first, the weak form of Eq. (3) is obtained.

$$\begin{aligned} &\int_0^L \int_{R_i}^{R_o} \left[\mathrm{K}\frac{\partial(\delta T)}{\partial r}\frac{\partial T}{\partial r} + \mathrm{K}\frac{\partial(\delta T)}{\partial z}\frac{\partial T}{\partial z} + \rho c\left(\frac{\partial T}{\partial t} + \tau\frac{\partial^2 T}{\partial t^2}\right)\delta T \right] r dr dz \\ &\quad + \int_0^L (r q_r \delta T)\big|_{r=R_i}^{R_o} dz + \int_{R_i}^{R_o} (q_z \delta T)\big|_{z=0}^{L} r dr = 0 \end{aligned} \tag{7}$$

where δT and $q_n \left(= -K\frac{\partial T}{\partial n}\right)$ are the variation of temperature and the heat flux at an arbitrary material point of the cylinder.

The FG cylinder is divided into N_e elements along the z-direction. In each element, the weak form of the governing equation, i.e., Eq. (7), should be satisfied. In order to avoid the assembling procedure and consequently to reduce the computational effort of the FEM procedure, the global interpolation functions are used to approximate the temperature distribution within each element.

$$T(r, z, t) = \sum_{i=1}^{N_z} \hat{T}_i(r, t)\, \phi_i(z) \tag{8}$$

where N_z is the total number of nodes along the z-direction and $\phi_i(z)$ $(i = 1, 2, \ldots, N_z)$ are the global Lagrange interpolation functions, which for linear elements take the following form.

$$\phi_i(z) = \begin{cases} 0 & z \leq z_{i-1} \\ \frac{z - z_{i-1}}{z_i - z_{i-1}} & \mathrm{z}_{i-1} \leq z \leq z_i \\ \frac{z_{i+1} - z}{z_{i+1} - z_i} & z_i \leq z \leq z_{i+1} \\ 0 & z_{i+1} \leq z \end{cases} \quad \text{for } i = 1, 2, \ldots, N_z \tag{9}$$

Substituting the approximation presented in Eq. (8) into Eq. (7) and performing the integration over the length of the cylinder, one gets the following.

$$\int_{R_i}^{R_o}\left[\sum_{i=1}^{N_z}\sum_{j=1}^{N_z}\left[-\frac{A_{ij}}{r}\frac{\partial}{\partial r}\left(r\mathrm{K}(r)\frac{\partial\hat{T}_j}{\partial r}\right)+D_{ij}\mathrm{K}(r)\hat{T}_j+A_{ij}\rho(r)c(r)\left(\frac{\partial\hat{T}_j}{\partial t}+\tau\frac{\partial^2\hat{T}_j}{\partial t^2}\right)\right]\right.$$
$$\left.\delta\hat{T}_i r dr+\left\{r\sum_{i=1}^{N_z}\left[\int_0^L\phi_i\, q_r dz+K\sum_{j=1}^{N_z}A_{ij}\frac{\partial\hat{T}_j}{\partial r}\right]\delta\hat{T}_i\right\}\right|_{r=R_i}^{R_o}$$
$$+\int_{R_i}^{R_o}\left(\sum_{i=1}^{N_z}\phi_i\,\hat{q}_z\delta\hat{T}_i\right)\Bigg|_{z=0}^{L} r dr=0 \tag{10}$$

where

$$A_{ij}=\int_0^L\phi_i\phi_j dz,\; D_{ij}=\int_0^L\frac{d\phi_i}{dz}\frac{d\phi_j}{dz}dz \tag{11a, 11b}$$

Using the linear independence of the terms $\delta\hat{T}_i$ ($i=1, 2, \ldots, N_z$) and after implementing the boundary conditions at the surfaces $z=0$ and L, the governing differential equation, and the related boundary conditions at each FE nodal line i with $i=1, 2, \ldots, N_z$ are obtained as follows,

$$\sum_{j=1}^{N_z}\left[-\frac{A_{ij}}{r}\frac{\partial}{\partial r}\left(r\mathrm{K}(r)\frac{\partial\hat{T}_j}{\partial r}\right)+D_{ij}\mathrm{K}(r)\hat{T}_j+A_{ij}\rho(r)c(r)\left(\frac{\partial\hat{T}_j}{\partial t}+\tau\frac{\partial^2\hat{T}_j}{\partial t^2}\right)\right]=0 \tag{12}$$

$$\text{At } r=R_i:\; K\sum_{j=1}^{N_z}A_{ij}\frac{\partial\hat{T}_j}{\partial r}=f_i(t);\quad \text{at } r=R_o:\;\sum_{j=1}^{N_z}\left(KA_{ij}\frac{\partial\hat{T}_j}{\partial r}+\hat{A}_{ij}\hat{T}_j\right)=g_i(t) \tag{13a, 13b}$$

where

$$f_i(t)=-\int_0^L\phi_i\,\hat{q}_r dz,\;\hat{A}_{ij}=\int_0^L h_c\phi_i\phi_j\, dz,\; g_i(t)=\int_0^L h_c T_\infty\phi_i\phi_j\, dz \tag{14a – 14c}$$

It is obvious that the resulting governing differential equations have variable coefficients and consequently, if it is not impossible to solve them analytically it is very difficult to obtain such a solution. Hence, here the differential quadrature method (DQM) as an efficient and accurate numerical tool [17, 29–34] in conjunction with the Newmark time integration scheme are employed to discretize this system of equations in the radial direction and temporal domain, respectively. A brief review of DQM is presented in Appendix A. Based on the DQM, at each finite element nodal line i, the cylinder is discretized into a set of N_r discrete grid points along the r-direction. Then, at each domain grid point r_k with $k=2,3,\ldots,N_r-1$, the heat transfer Eq. (12) can be discretized as follows.

$$\sum_{j=1}^{N_z}\left\{-\sum_{m=1}^{N_r}A_{ij}\left[\left(\frac{\mathrm{K}_k}{r_k}+\left(\frac{d\mathrm{K}}{dr}\right)_k\right)A^r_{km}+\mathrm{K}_kB^r_{km}\right]\hat{T}_{jm}+D_{ij}\mathrm{K}_k\hat{T}_{jk}\right.$$
$$\left.+A_{ij}\rho_kc_k\left(\frac{d\hat{T}_{jk}}{dt}+\tau\frac{d^2\hat{T}_{jk}}{dt^2}\right)\right\}=0 \tag{15}$$

where A^r_{ij} and B^r_{ij} represent the weighting coefficients of the first and second-order derivatives along the r-direction, respectively [33]; also $(\ )_k$ means the function value at $r=r_k$.

In a similar manner, the DQ-analogs of the initial and boundary conditions can be represented as follows.

Equation (4a, 4b):

$$\hat{T}_{jk}(0)=T_0,\left.\frac{d\hat{T}_{jk}}{dt}\right|_{t=0}=0 \tag{16a, 16b}$$

Equation (13a):

$$K_k\sum_{j=1}^{N_z}\sum_{m=1}^{N_r}A_{ij}A^r_{km}\hat{T}_{jm}=f_i(t) \tag{17a}$$

Equation (13b):

$$\sum_{j=1}^{N_z}\left(K_kA_{ij}\sum_{m=1}^{N_r}A^r_{km}\hat{T}_{jm}+\hat{A}_{ij}\hat{T}_{jk}\right)=g_i(t) \tag{17b}$$

In the matrix form, the discretized governing differential equation and the associated boundary conditions can be represented as follows.

$$[M]\left\{\frac{d^2\hat{T}}{dt^2}\right\}+[C]\left\{\frac{d\hat{T}}{dt}\right\}+[K]\{\hat{T}\}=\{F\} \tag{18}$$

where $\{\hat{T}\}$ is the vector of degrees of freedom and based on the arrangement of its elements, the elements of the coefficient matrixes $[M]$, $[C]$, $[K]$ and the load vector $\{F\}$ are obtained from the discretized form of the governing differential equation (15) and the related boundary conditions (17).

Using the Newmark time integration scheme [35], the system of differential Eq. (18) is transformed into a set of linear algebraic equations. After solving it, the temperature field at each time step and at different spatial grid points can be obtained. At the end of each time step, the temperature is used as the initial condition for the next time step.

Table 1. Material properties of ceramic (zirconia) and metals (steel) [6]

Material	k (W/cm°C)	ρ (kg/m^3)	c (J/kg°C)
Zirconia	2.09	5700	531.9
Steel	20	8166	325.35

4. NUMERICAL RESULTS

In this section, first, the convergence behavior of the method for evaluating the temperature distribution of an FG hollow cylinder under a heat flux with moving

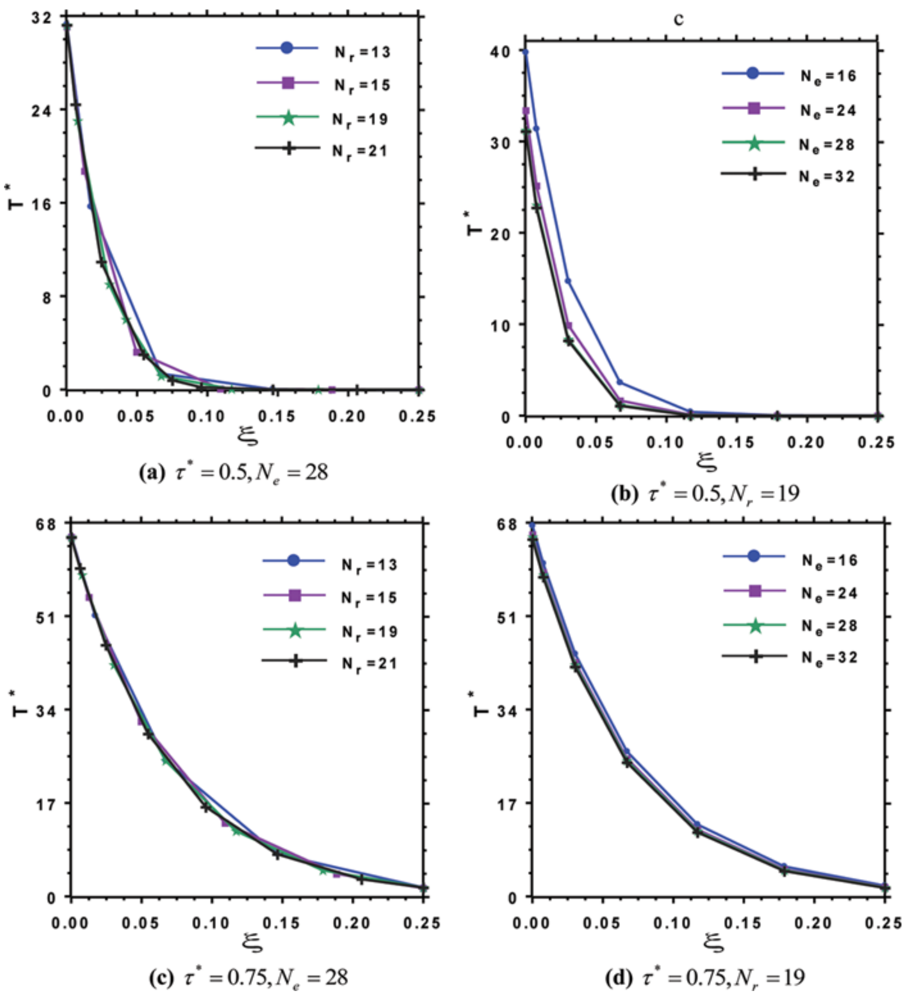

Figure 2. Convergence of the nondimensional temperature of a hollow FG cylinder subjected to distributed heat flux with a moving front edge ($p = 1$, $V_e = 0$, $\eta = 0.5$). (*a*) $\tau^* = 0.5$, $N_e = 28$, (*b*) $\tau^* = 0.5$, $N_r = 19$, (*c*) $\tau^* = 0.75$, $N_e = 28$, and (*d*) $\tau^* = 0.75$, $N_r = 19$ (color figure available online).

front edge is investigated. Without loss of generality, in all problems considered here equal length elements are used. The FG material properties are assumed to vary according to power law distribution (1); also, the nondimensional parameters are defined as follows.

$$\eta = z/L,\ \xi = \frac{r - R_i}{R_o - R_i},\ T^* = \frac{h_c T L^2}{q_0 h^2},\ \tau^* = \frac{ut}{L},\ V_e = \sqrt{\frac{\alpha_c \tau}{h^2}} \tag{19}$$

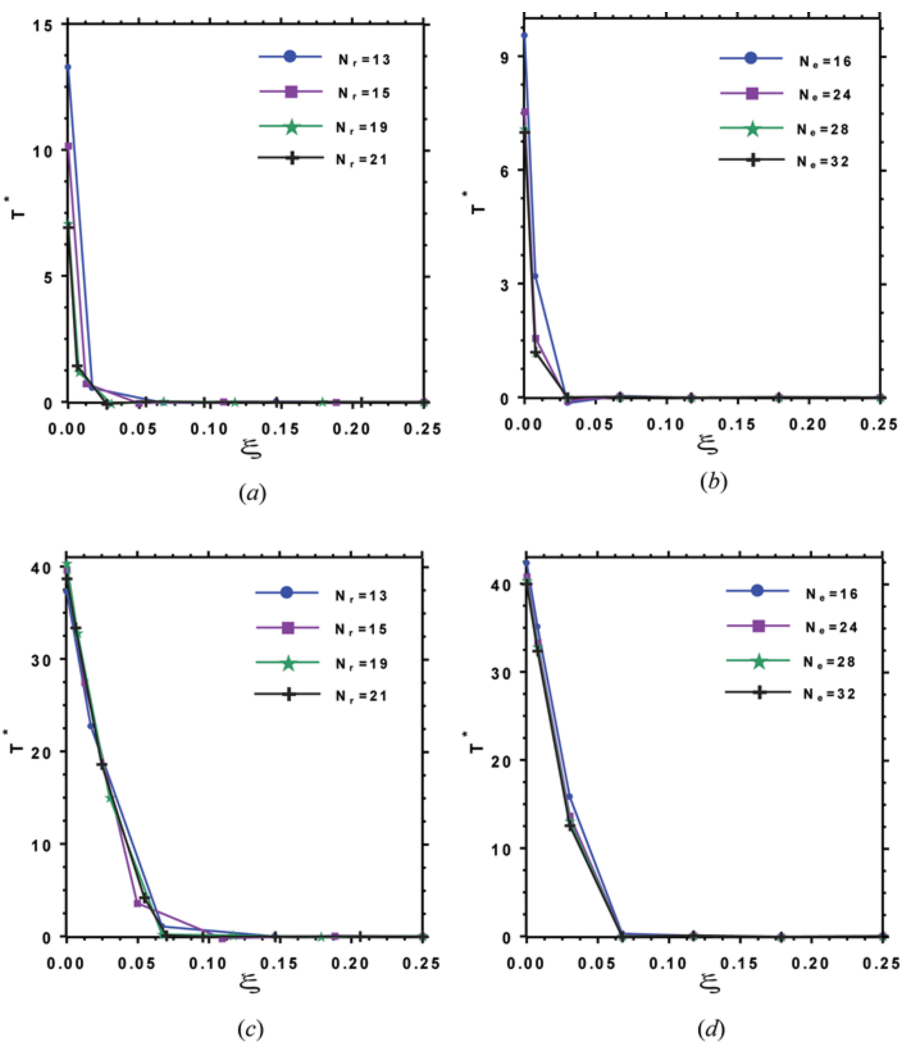

Figure 3. Convergence of the nondimensional temperature of a hollow FG cylinder subjected to distributed heat flux with a moving front edge ($p = 1$, $V_e = 0.1$, $\eta = 0.5$). (*a*) $\tau^* = 0.5$, $N_e = 28$, (*b*) $\tau^* = 0.5$, $N_r = 19$, (*c*) $\tau^* = 0.75$, $N_e = 28$, and (*d*) $\tau^* = 0.75$, $N_r = 19$ (color figure available online).

where α_c $(=K_c/\rho_c c_c)$is the thermal diffusivity of the ceramic, $h(=R_o-R_i)$ is the thickness, and L the length of the hollow cylinder; u is the velocity of heat flux front edge, which is assumed to be constant. The effect of relaxation time τ appears in the Vernotte number (V_e)and it can be seen that by increasing the Vernotte number, the non-Fourier effect increases. Also, it is obvious that when $V_e=0$, the results become equal to those obtained using the Fourier model. Unless otherwise specified, the following values of different parameters are used in the solved examples.

$$u=2(\mathrm{cm/s}),\ q_0=200\left(\mathrm{W/cm^2}\right),\ h_c=10\left(\mathrm{W/cm^2\,{}^oC}\right), R_i=4\ (\mathrm{cm}),$$
$$R_o=6\ (\mathrm{cm}),\ L=20\ (\mathrm{cm}) \tag{20}$$

The material properties of zirconia and steel, as given in Table 1, are used to obtain the new numerical results, which together with the abovementioned geometrical parameters are chosen from the work of Santos et al. [6].

As a first example, the convergence behavior of the nondimensional temperature across the thickness of an FG cylinder against the number of DQ grid points along the r-direction and the number of elements along the z-direction based on both Fourier and hyperbolic heat conduction models are shown in Figures 2 and 3, respectively. It is noticeable that by increasing the number of DQ grid points and elements, the results converge without any numerical instability.

Due to a lack of numerical results for an FG hollow cylinder subjected to an axially moving front edge heat flux, the result of an FG thick-walled hollow cylinder subjected to a prescribed transient temperature on its inner surface obtained by

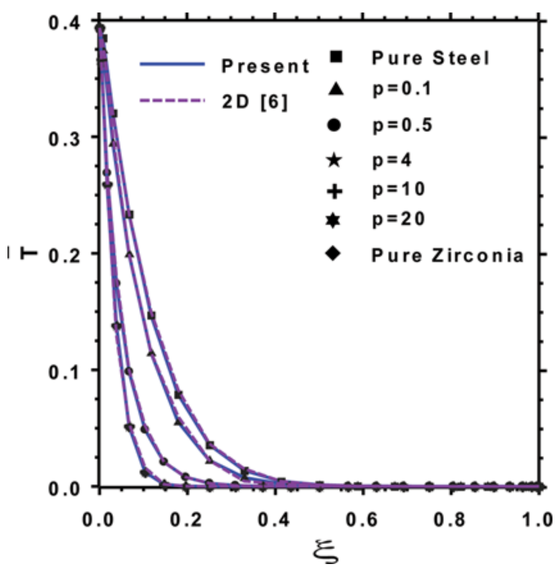

Figure 4. Comparison of the transient nondimensional temperature $\left(\overline{T}=T/\overline{T}_0\right)$ distributions across the thickness of an FG hollow cylinder subjected to prescribed temperature on its inner surface (color figure available online).

Santos et al. [6] is used to verify the presented approach. The boundary conditions at different boundaries of the FG hollow cylinder are as follows [6].

$$\begin{aligned}&\text{At } r = R_i : T(r,z,t) = \overline{T}_0\left(1 - e^{-0.5t}\right);\ \text{at } r = R_o : K\frac{\partial T}{\partial r} + h_c T = 0;\\ &\quad\text{at } z = 0,\ L : T(r,z,t) = 0;\ T(r,z,0) = 0\end{aligned} \tag{21}$$

where the geometrical and other parameters are defined in Eq. (20) and Table 1. A semi-analytical finite element model was employed by Santos et al. [6] to solve this

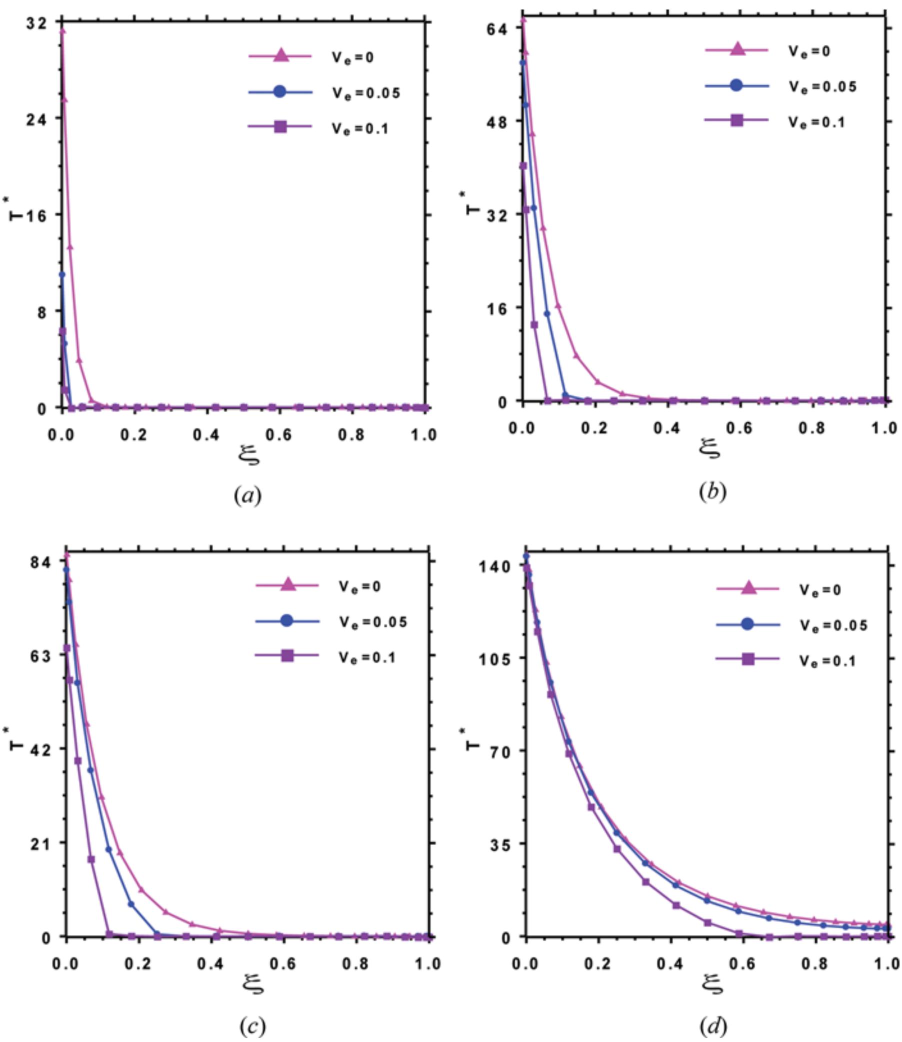

Figure 5. Influences of the Vernotte number on the nondimensional transient temperature for the FG hollow cylinder at various time levels ($N_e = 28$, $N_r = 19$, $p = 1$, $\eta = 0.5$). (*a*) $\tau^* = 0.5$, (*b*) $\tau^* = 0.75$, (*c*) $\tau^* = 1$, and (*d*) $\tau^* = 2.5$ (color figure available online).

problem. All the material properties were assumed to vary according to power law distribution Eq. (1). The temperature distribution across the thickness of the FG hollow cylinder for different values of the material graded index (p) is compared with those of Santos et al. [6] in Figure 4. Excellent agreement between the results of the two approaches is evident. In addition, it is obvious that by increasing the material graded index (p), the results approach to those of a shell fully made of zirconia.

After demonstrating the convergence and accuracy of the method, some parametric studies for FG hollow cylinders subjected to heat flux with a moving front edge are performed.

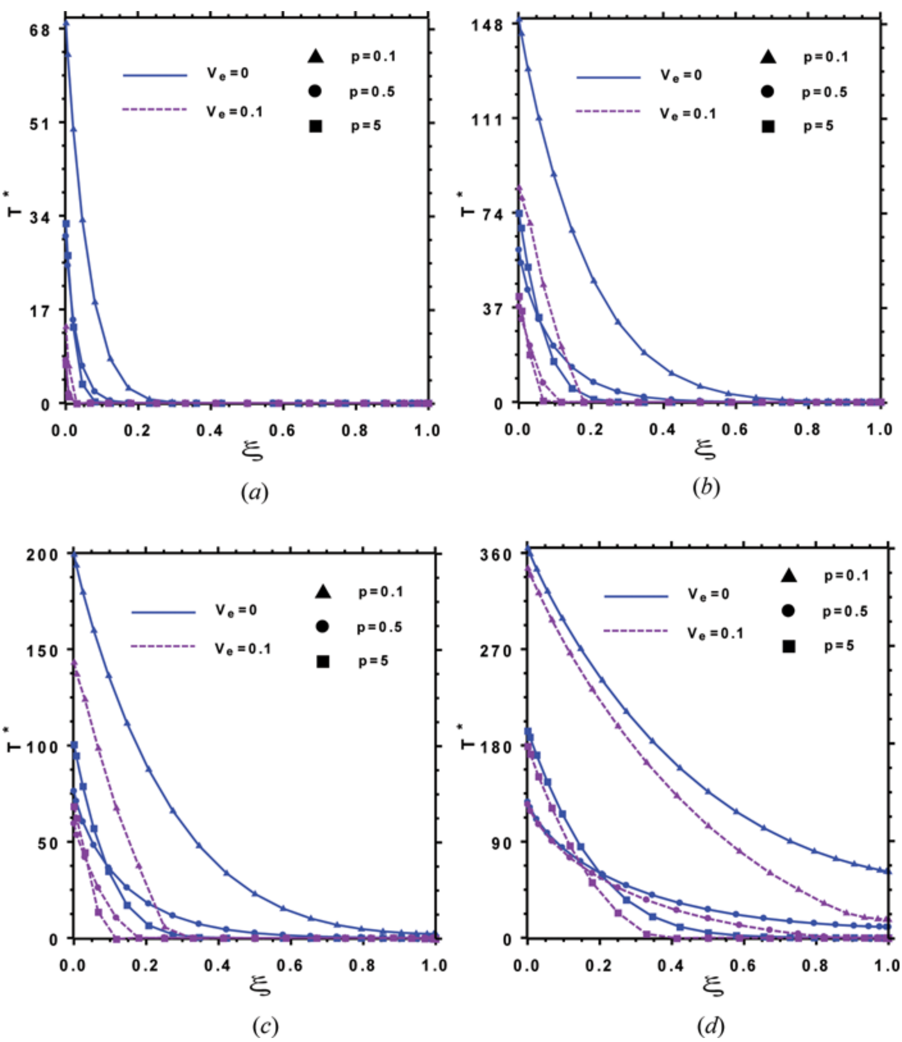

Figure 6. Influences of the power law index on the nondimensional transient temperature for the FG hollow cylinder at various time levels ($N_e = 28$, $N_r = 19$, $\eta = 0.5$). (*a*) $\tau^* = 0.5$, (*b*) $\tau^* = 0.75$, (*c*) $\tau^* = 1$, and (*d*) $\tau^* = 2.5$ (color figure available online).

As a first example, the effect of the Vernotte number (V_e) on the nondimensional temperature distribution across the thickness of FG cylinder at various time levels are shown in Figure 5. It is clear that by increasing the Vernotte number, the non-Fourier effect increases and, consequently, the nondimensional temperature of the points far from the inner surface decreases.

Comparison between the results of the Fourier and hyperbolic models is seen in Figure 6. The effect of different values of the material graded index (p) on the temperature distribution at different time levels are shown. It can be seen that the material graded index significantly affects the nondimensional temperature distribution across the thickness of the cylinder, and also the non-Fourier effects cannot be neglected.

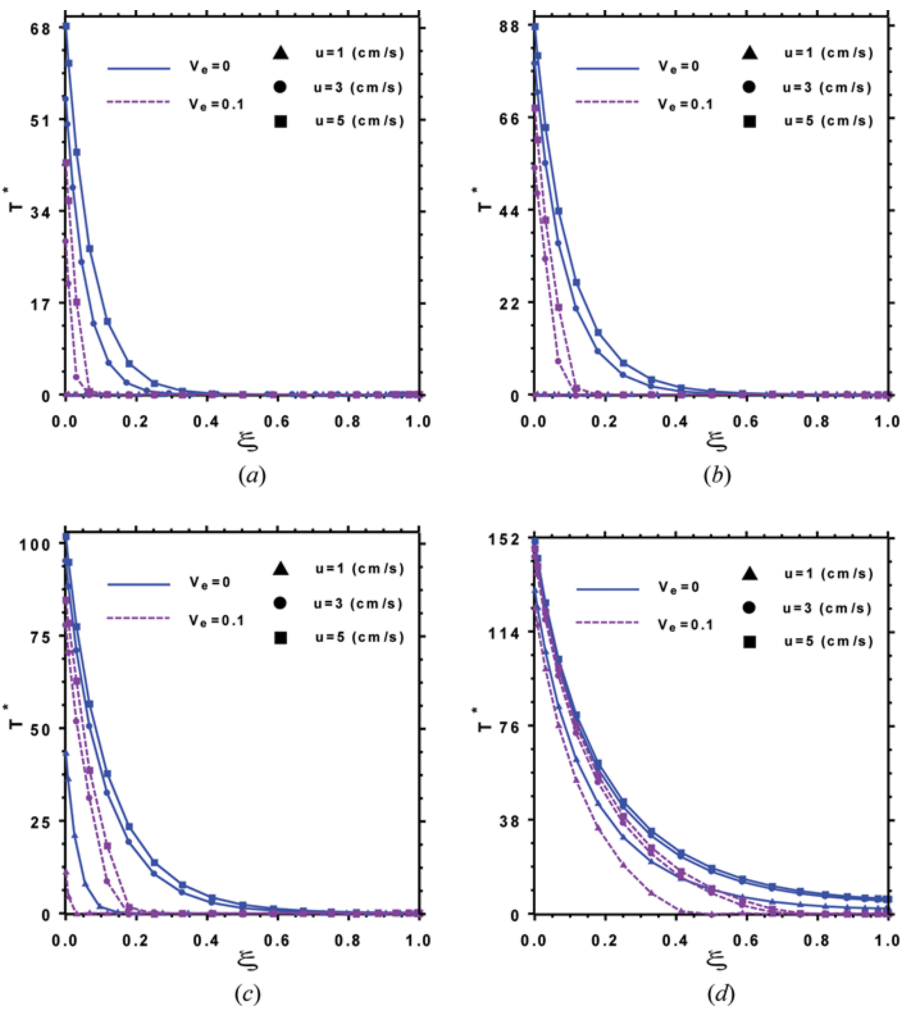

Figure 7. Influences of velocity of the moving edge of the heat flux on the nondimensional transient temperature for the FG hollow cylinder at various time levels ($N_e = 28$, $N_r = 19$, $p = 1$, $\eta = 0.5$). (*a*) $t = 5$ s, (*b*) $t = 7.5$ s, (*c*) $t = 10$ s, and (*d*) $t = 25$ s (color figure available online).

The effect of velocity of the heat flux front edge on the nondimensional temperature distribution across the thickness of the FG cylinder for various time levels based on Fourier and hyperbolic models are exhibited in Figure 7. It can be seen that increasing the velocity of the heat flux front edge, the nondimensional temperature at all points increases for all values of time levels. In addition, with the increase of elapsed time, the difference between the results of Fourier and hyperbolic models reduces.

Presented in Figure 8 are the time histories of the temperature at various points along the length and cross-section of the FG cylinder based on the both heat conduction models. It is noticeable that through the thickness effect of the hyperbolic model is greater than its effect along the longitudinal direction. Moreover, the non-Fourier effect decreases with increasing elapsed time.

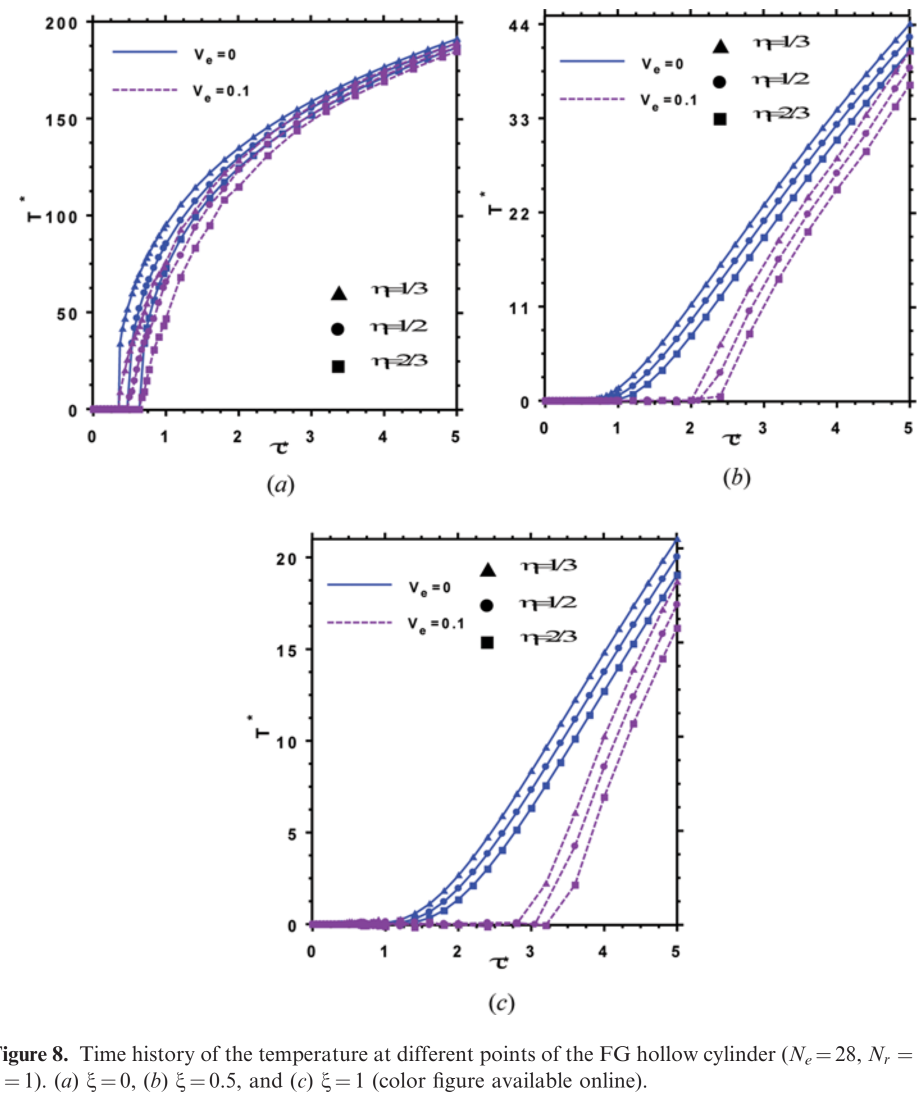

Figure 8. Time history of the temperature at different points of the FG hollow cylinder ($N_e = 28$, $N_r = 21$, $p = 1$). (*a*) $\xi = 0$, (*b*) $\xi = 0.5$, and (*c*) $\xi = 1$ (color figure available online).

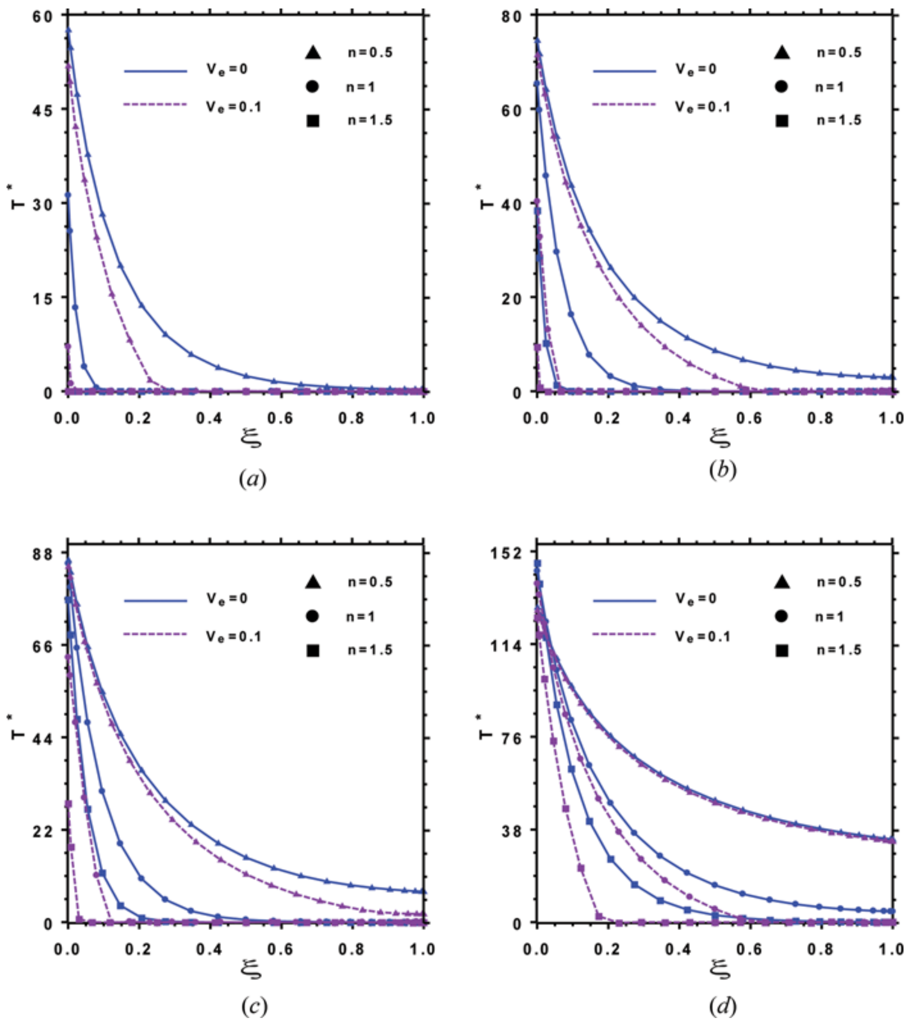

Figure 9. Influences of geometrical parameters on the nondimensional transient temperature for the FG hollow cylinder at various times $(N_e = 28, N_r = 21, h = 2n(\text{cm}), L = 20n(\text{cm}), p = 1, \eta = 0.5)$. (*a*) $t = 5$ s, (*b*) $t = 7.5$ s, (*c*) $t = 10$ s, and (*d*) $t = 25$ s (color figure available online).

The influences of thickness and length of the FG cylinder on the nondimensional temperature distribution across its thickness based on the both heat conduction models are presented in Figure 9. It can be seen in this figure that by increasing the thickness and length of the cylinder, the non dimensional temperature reduces. In addition, this influence increases when the elapsed time increases.

5. CONCLUSION

The heat transfer analysis of FG hollow cylinders subjected to heat flux with a moving front edge on their inner surface and convective heat transfer on their outer surface is presented. In order to include the non-Fourier effect, the hyperbolic heat transfer equations are used. The material properties are assumed to be graded in the thickness direction. The governing differential equations are discretized in the longitudinal direction of the FG cylinder using the FEM. Then, the resulting system of differential equations and the related boundary conditions are discretized in the radial direction and in their strong form by using DQM. The DQ formulations precisely satisfy the boundary conditions on the inner and outer surfaces of the hollow cylinder. The resulting system of ordinary differential equations is solved in temporal domain using the Newmark time integration scheme. The convergence behavior of the method is numerically demonstrated and comparison studies with the available solutions in the literature are performed to validate the presented formulation and the method of solution. Parametric studies are performed to study the thermal behavior of FG hollow cylinders subjected to heat flux with a moving front edge. From the obtained results, it is concluded that the material graded index, the Vernotte number, the geometrical parameters, and the velocity of the heat source have significant effects on the temperature distribution of an FG hollow cylinders. In addition, it is concluded that the non-Fourier effect cannot be neglected for an accurate heat transfer analysis of an FG cylinder under transient thermal loading.

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APPENDIX A. A BRIEF REVIEW OF DQM

The basic idea of the differential quadrature method is that the derivative of a function, with respect to a space variable at a given sampling point, is approximated as the linear weighted sums of its values at all of the sampling points in the domain of that variable. In order to illustrate the DQ approximation, consider a function $f(r,\ t)$ having its field on a domain $R_i \leq r \leq R_o$ and $0 \leq t$.Let, in the given domain, the function values be known or desired on a grid of sampling point. According to the DQ method, the first and second-order derivatives of the function $f(r,\ t)$ can be approximated as follows

$$\left.\frac{\partial f(r,t)}{\partial r}\right|_{r=r_i} = \sum_{m=1}^{N_r} A_{im}^r f(r_m,t) = \sum_{m=1}^{N_r} A_{im}^r f_m;\ \left.\frac{\partial^2 f(r,r)}{\partial r^2}\right|_{r=r_i} = \sum_{m=1}^{N_r} B_{im}^r f_m \text{ for } i=1,2,\ldots,N_r \tag{A1}$$

From this equation one can deduce that the important components of DQ approximations are weighting coefficients and the choice of sampling points. In order to determine the weighting coefficients, a set of test functions should be used in Eq. (A1). For polynomial basis functions DQ, a set of Lagrange polynomials are employed as the test functions. The weighting coefficients for the first-order derivatives in ξ-direction are thus determined as follows [29].

$$A_{ij}^r = \begin{cases} \frac{1}{(R_o-R_i)}\frac{M(r_i)}{(r_i-r_j)M(r_j)} & \text{for } i \neq j \\ -\sum\limits_{\substack{j=1\\ i\neq j}}^{N_r} A_{ij}^r & \text{for } i=j \end{cases} \quad i,j=1,2\ldots,N_r \tag{A2}$$

where $M(r_i) = \prod_{k=1, i\neq k}^{N_r} (r_i - r_k)$.

The weighting coefficients of second order derivative can be obtained as follows [29].

$$[B^r_{ij}] = [A^r_{ij}][A^r_{ij}] = [A^r_{ij}]^2 \tag{A3}$$

In numerical computations, Chebyshev-Gauss-Lobatto quadrature points are used, that is [29],

$$r_k = R_i + \frac{(R_o - R_i)}{2}\left\{1 - \cos\left[\frac{(k-1)\pi}{(N_r - 1)}\right]\right\} \textit{ for } k = 1, 2, \ldots, N_r \tag{A4}$$